

Introduction: Coronary Artery Disease (CAD) is a type of heart disease that occurs when there is too much plaque buildup in the arteries, leading to the heart not being able to deliver enough oxygenated blood to the rest of the body. About 1 in 20 adults over 20 have CAD, making it the most common type of heart disease; In the US in 2022 alone, it has caused 371,506 deaths. CAD falls under three categories: obstructive CAD (in which 50% or more of a large artery is blocked), non obstructive CAD (in which less than 50% of a large artery is blocked), and coronary microvascular disease (in which the tiny arteries in the heart are blocked) [1]. The risk factors for CAD include high bad cholesterol (LDL), low good cholesterol (HDL), high blood pressure, family history, diabetes, smoking, and obesity [2]. In general, CAD is preventable with a good diet and exercise, and if patients take medication, it's entirely possible to stop or even reverse the effects of plaque buildup [1]. The goal of this project is to classify patients into CAD and non CAD categories using an SVM.

Study Design: The data was downloaded from the UCI data repository, featuring datasets from Cleveland, Hungary, Switzerland, and VA Long Beach [3]. The Cleveland dataset consists of 303 patients, with a mean age of 54, who were referred for coronary angiography between 1981 and 1984. None of the patients had any prior evidence of heart attacks or cardiomyopathic disease. Their history was taken, along with a physical exam, ecg, serum cholesterol test, and fasting blood sugar. After consent was given, they also got an exercise ecg, thallium scintigraphy, and cardiac fluoroscopy. 200 patients were studied at the Veterans Administration Medical Center in Long Beach, 294 patients at the Hungarian Institute of Cardiology, and 123 patients across two university hospitals in Switzerland. For these three datasets, noninvasive tests and coronary angiography were taken for the patients. Histories and physical exams were given within 6 weeks before the coronary angiogram was given. If there was a greater than 50% narrowing of a major vessel, then the results for the coronary angiogram was considered abnormal [4]. All the datasets contain the columns age, sex (0: female, 1: male), chest pain (1: typical angina, 2: atypical angina, 3: non-anginal pain, 4: asymptomatic), resting blood pressure (resttbps), serum cholesterol (chol), fasting blood sugar (fbs), resting ecg (restecg with 0: normal, 1: ST-T wave abnormality, 2: left ventricle hypertrophy), maximum heart rate (thalach), exercise induced angina (exang, with 0: no, 1: yes), ST depression induced by exercise (oldpeak), slope of the peak exercise ST segment (slope, 1: upsloping, 2: flat, 3: downsloping), number of major vessels, 0 to 3, colored by fluoroscopy (ca), thallium scintigraphy results (thal, 3: normal, 6: fixed defect, 7: reversible defect), and diagnosis of heart disease (num, 0: <50% diameter narrowing, 1: >50% diameter narrowing) [3].

SVM: A Support Vector Machine (SVM) is a machine learning algorithm that uses supervised learning to find a boundary line or plane that maximizes the distance between classes, based on the support vectors, or the points closest to the boundary [7]. They are useful as they can handle classifications that are both linear and non-linear. The number of variables used in this model dictates whether the boundary, or hyperplane, will be a line, a plane, or something else. When the data cannot be separated in a linear fashion, a kernel function can be used to model the data in a higher dimensional space where a boundary can be created with more ease than before [5][6]. The type of kernel used depends on the type of SVM one wants to use. The basic equation is $wx + b = 0$, in which w represents the weights, or the effect of each variable on the

boundary, and b represents the bias, or the movement of the boundary. In this case, I am attempting to predict CAD status in a patient, so a positive value will indicate CAD presence, a negative value will indicate no CAD presence, and a value of 1 will indicate that the point rests on the boundary itself. To work with a linear SVM, which I used in this project, the assumptions that need to be met are that the data can be reasonably separated using a line, that the variables are all on the same scale, and that the data points are independent.

Data Analysis: The first step was to combine the four datasets into one large dataset, and add a site variable to account for any baseline CAD rate differences. Then I checked what the amount of missing values are in the dataset. CA and THAL had the most amount of missing variables, so I decided to remove them from the analysis. For the rest of the values, I imputed using the median, as it deals with outliers better.

age	0.000000
sex	0.000000
cp	0.000000
trestbps	6.413043
chol	3.260870
fbs	9.782609
restecg	0.217391
thalach	5.978261
exang	5.978261
oldpeak	6.739130
slope	33.586957
ca	66.413043
thal	52.826087
target	0.000000
source	0.000000

Figure 1: Percentage of Missing Values by Variable

I then made the CAD variable a binary variable, and converted it to +1 and -1, as that's what SVM calculations require. The data was then split; 80% went into the training group and 20% went into the testing group. The variables of interest were then scaled, such that they would have a mean of 0, and a standard deviation of 1. This ensures that all the variables will be on the same scale for when gradient descent is calculated, as it updates all weights by the same learning rate. If this isn't done, then some variables with higher values purely based on the numbers, such as cholesterol, will dominate variables with lower values. To set up the SVM, I initialized the variables C , $learning_rate$, and n_epochs . C , representing the tradeoff between correctly classifying every value and a wide margin, was set to 1. $learning_rate$, or how big of a step each update takes, was set to 0.001, and n_epochs , the number of times the training dataset is looped over, was set to 1000. I then calculated λ , which is equal to $1/C$. And

before I created the actual SVM calculation, I initialized a vector of zeros representing weight, 1 per feature, and the bias, which was set at 0. From there, I proceeded to the gradient descent. For each epoch, each patient's feature values were matrix multiplied by the weights, and summed. Then I added the bias and multiplied by y, flipping the sign to negative if the guess was wrong. So the equation became $y * (\text{sum}(\text{values} * \text{weights}) + \text{self.b})$. A true/false vector was then created, with the true meaning that the prediction was either wrong or just not confident enough (the number wasn't as large as it should be), and those values became the learning points for the next epoch. The loss function, representing how wrong the model is at any point, is as follows:

$L = (\lambda/2)||w||^2 + (1/n) \sum \max(0, 1 - y_i(w \cdot x_i + b))$, where $(\lambda/2)||w||^2$ is the weights being squared and summed, and then scaled by lambda to ensure the steps are small. $(1/n) \sum$ represents the average across all patients. In $\max(0, 1 - y_i(w \cdot x_i + b))$, $w \cdot x_i + b$ is the raw score per patient, and y_i is the +1 or -1 label for CAD. This whole thing is then subtracted from 1 so that a negative output will represent a correct answer, and a positive output will represent an incorrect answer. The $\max(0, \dots)$ tells the computer to take whichever is higher, 0 or the number calculated just then. The goal is to minimize this loss, which is where the gradients come in. Getting the derivative of the equation in relation to w and b gets the following:

If the patient is correctly identified confidently, then the gradient is $dL/dw = \lambda w$, and b's gradient is $dL/db = 0$. If the identification is wrong or not confident, then the loss is taken into account and the gradient is $dL/dw = \lambda w - y_i x_i$, and b's gradient is $dL/db = -y_i$. To update the model with this new information, I used the following equations:

$w = w - \text{learning_rate} \times dL/dw$

$b = b - \text{learning_rate} \times dL/db$

The reason I subtracted the learning rate is because the gradient is pointing uphill, and each step taken has to go downhill to minimize the loss. All this is then used to make the decision, by solving $w \cdot x + b$, where w is the weight, x is the patient and all its data, and b is the bias. The prediction just depends on the sign of the outcome, where positive indicates CAD presence, and negative indicates CAD absence. The final step in the analysis was to create the SVM itself, which was done by `svm = SVM(C=1.0, learning_rate=0.01, n_epochs=2000)`. The output after fitting the training and testing datasets, and then predicting were the AUC values, which represent the percent guessed correctly. 0.5 indicates random guessing, so the aim is to get higher than that.

Accuracy: 0.810				
ROC-AUC: 0.888				
	precision	recall	f1-score	support
No CAD	0.81	0.76	0.78	82
CAD	0.81	0.85	0.83	102
accuracy			0.81	184
macro avg	0.81	0.80	0.81	184
weighted avg	0.81	0.81	0.81	184

Figure 2: Output

The accuracy for this model was 0.81, meaning that 81% of patients were classified correctly. The ROC-AUC value states that if one random CAD patient, and one random non-CAD patient were taken and compared, the model will rate the risk of the CAD patient as higher 88% of the time. The precision says that 81% of the time, when a patient is CAD+, the model will identify it correctly, and same with non-CAD patients. The recall is the amount of patients correctly identified out of the total patients in that group. So out of all non-CAD patients, the model caught 76% of them, and out of the CAD patients, 85%. F1 just combines the precision and recall values with an equation. Support just tells, in each class, how many of each group are there.

References:

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